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An improved facial emotion recognition system using convolutional neural network for the optimization of human robot interaction

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Artificial intelligence (AI) has been effectively augmenting the features of robotics applications, including surveillance, medical support, aid services for the elderly or disabled, and many more uses. Most robotics applications need a variety of human-robot interactions (HRI) for effective and accurate use. Computer vision plays a vital role in improved and precise HRI. The human facial emotion recognition (FER) technique is one of the important types of computer vision that is crucial for the enhancement of HRI. This paper presents a study that uses algorithms based on computer vision and machine learning (ML) to identify the emotional states of humans in photographs and videos while users interact with various visual objects. Using ML techniques and a digital image processing pipeline, this study aims to reveal the development of software that can recognize emotions in human gestures. This research shows that software creation and training based on emotional expressions, together with the use of convolutional neural networks (CNNs) for emotional identification, are feasible. The main objective of this study is to find the essential facial gestures that the CNN framework with FER emphasizes. Finally, the proposed model is comparatively analyzed against FER2013, the Real-world Affective Faces Database (RAF-DB), and CK+ datasets. The rate of accuracy is highest for the CK+ dataset, which is around 95% and lowest for the FER2013 dataset, which is around 64%. The findings of this research will advance our knowledge about neural networks and help enhance the efficiency of computer vision.

An interdisciplinary topic called "affective computing" studies and develops computers that can understand and analyze human emotions[1]. Facial expressions are an important source of non-verbal cues in human interaction that might lend depth and significance to spoken words. According to certain research, 60–80% of communication takes place nonverbally[2]. These non-verbal cues include physical distance, hand gestures, quality of voice, eye contact, and facial expressions. In the domain of human-robot interaction (HRI), facial emotion recognition (FER) is used in surveillance, robotic navigation, education, human-activity recognition (HAR), medical care, psychological care, autopilot, and behavioral research in computer vision[3,4]. Mobile robots are autonomous devices that use preprogramming to see, identify, work together, and perform a variety of duties, including manufacturing activities, medical support, customer service to consumers, security and civil surveillance, and more[5,6]. The ability to simulate empathy for various controllers is required[7], that is, to give robots the ability to identify, understand, and therefore produce adaptable actions using recognized information, which is a major driving force behind studies in this domain.

Artificial intelligence (AI) technologies have made a significant contribution to developing human awareness, consciousness, actions, behavior, and learning[8]. Future advancements of AI depend on human-AI cooperation. In addition to fully autonomous or manually controlled equipment, robots can cooperate with a user companion with differing levels of automation and assistance[9]. Emotions are basic human characteristics that influence decisions and behavior[10,11]. They are vital to interaction, and effective relationships depend on emotional intelligence, that is, the capacity to recognize, utilize, and control the way one feels[12]. Human-like observation, determination, and expressive emotion skills are desired for robots in the HRI framework[13,14]. The ability of the robotic device to display recognized emotional expressions has a significant influence on social

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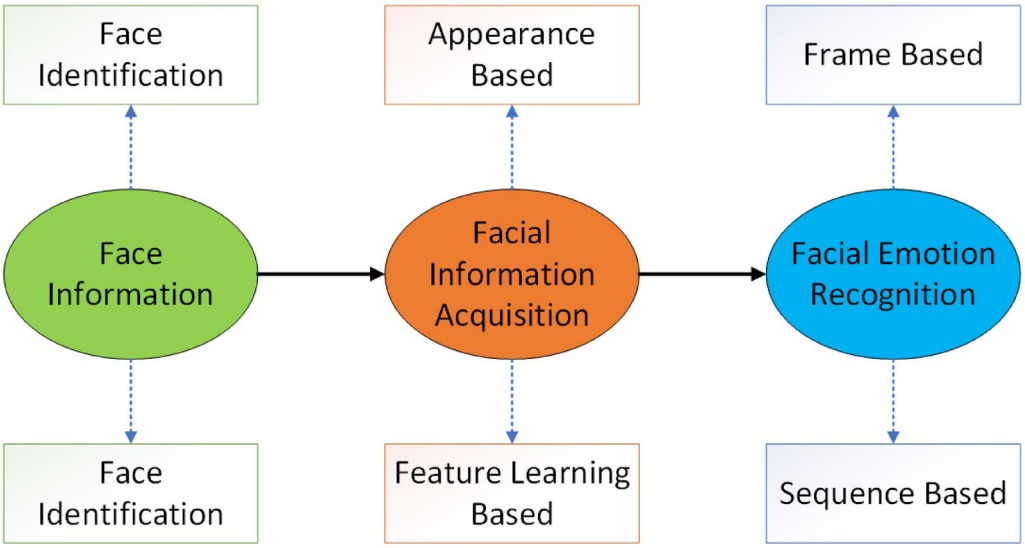

Fig. 1. Different types of techniques of the FER system.

Fig. 2. Illustration of the basic architecture of an FER system.

behavior that results in the application of complicated communication situations, including assistive, industrial, educational, and interpersonal robotics[15,16]. Several studies investigated how individuals perceive and identify states of consciousness, as well as which modalities, including human pose estimation[17,18], gesture recognition[19], HAR[20–22], and many more, can transmit psychological data from robots to humans. Figure 1 explains the different types of approach for FER systems.

Human emotion-interpreting and inferring robots would be more useful in social interactions. Face recognition and other machine learning algorithms might be applied to analyze emotion data and determine the mental state of a person. The face can transmit a variety of signals as well as gestures. When deciphered by computers, these delicate facial emotion signals can enhance the safety and coherence of HRI. Facial expressions can be considered a reliable indicator of a person's genuine feelings[23]. The idea behind psychological computing is to improve HRI by identifying the psychological conditions of the person using it.

The potential to provide robots with emotional intelligence with the goal of improving the intuitiveness, sincerity, and naturalness of HRI remains an interesting challenge within the subject. The robot's capacity to sense and comprehend human emotions is essential to achieving this. Numerous fields can benefit from the application of automated facial expression investigation, including the multimodal human-computer interface (HCI), psychological counseling, psychiatry, neuroscience, assessment of pain, recognition of lies, and interpersonal and linguistic communication. One of the most important non-verbal mechanisms that human-machine interface (HMI) technologies use to understand human intents and individual emotions is FER[24]. The FER scheme is considered a classification task. A collection of features generated from the initial image is fed into the classifier, which is shown in Fig. 2.

The feature extraction techniques that use static images are the Haar wavelet transform[25], Gabor wavelet transform[26], and Active Appearance Models (AAM)[27]. On the other hand, dynamic methods presume a temporal correlation in the sequencing of the expression of the input face throughout consecutive frames[28,29]. Several prevalent methods for recognizing emotions include artificial neural networks (ANN), AdaBoost, hidden Markov models, and support vector machines (SVM)[30]. Deep learning has made significant developments, and CNN's use has shown great promise[31,32]. However, a significant problem with deep learning is that usually developing good models requires a lot of data.

The algorithm developed by CNN has experienced considerable advancements in the identification of facial expressions, but there are still some shortcomings, such as excessively long training cycles and poor recognition rates in complicated environments. Two issues with ML breakthroughs in FER approaches have been noted in datasets that are now in use: (1) limited images and (2) images acquired during highly controlled settings. These issues led to the development of FER methods that prioritize the collection of web images[33]. The development of an intelligent, multimodal HRI framework that performs in a real-time environment is the primary goal of this

present work. The purpose of the present study is to improve the efficiency of the FER system by developing an automatic FER system that uses CNN.

There are many stages involved in facial recognition: image detection, image pre-processing, feature retrieval, image alignment, and image recognition[34]. Geometric characteristic extraction and total statistics characteristics-focused techniques are the two main methods for extracting features. For the purpose of characterizing the locations of the face organs as classification characteristics, the geometric feature-based technique is often employed[35]. The goal of this research is to provide a technique for building an FER scheme using CNN. Using a camera, the given model might be utilized in real time for FER. Examining and comprehending the benefits of deep CNN models over alternative deep learning models are analyzed.

The rest of the article is organized as follows: Section 2 provides the literature survey, which contains recent articles related to FER for HRI; Section 3 provides the background information required for the FER system; Section 4 presents the simulation and experimental analysis of the proposed methodology; Section 5 discusses the further details of this experimental study and future research directions; and finally, Section 6 discusses the concluding remarks on the proposed approach.

Related work

Automatic FER is still an extremely challenging and fascinating task in computer vision. DNNs are being used more and more to build discriminatory models for automatic FER as a result of the evolution of FER beyond controlled research environments, too difficult in wild situations, and the present popularity of ML techniques in a variety of domains. In recent decades, various researchers have focused on facial gesture recognition to enhance HRI.

Mollahosseini et al.[36] present a deep neural network (DNN) framework that can be used to solve the FER issue in several popular standard facial data sets. In particular, this network is composed of four Inception layers, two convolutional layers, and max pooling. The network, which consists of a single component design, classifies input photos of faces into either the neutral or the six fundamental expressions. In terms of accuracy and training time, the recommended design performs as well as or more effectively than state-of-the-art techniques and conventional convolutional neural networks (CNN).

Jung et al.[37] present a collaborative fine-tuning in DNNs for FER. The properties of temporal information are helpful in identifying facial expressions. However, it takes a lot of work to manually develop features that are beneficial. In this study, the deep learning approach is utilized, which is thought of as a method to automatically obtain important characteristics from unprocessed data. In order to minimize this effort. Two distinct models form the foundation of this deep network. One deep network uses imagery sequences to acquire temporal aspect features, and another uses temporal face landmark points to derive temporal geometrical information. Compared to conventional techniques such as feature synthesis and weighted summation, this new integrated technique yields more precise findings.

Kotsia et al.[38] present two new techniques for FER in face picture sequences. There are landmarks in the picture series that the user must personally configure part of the Candide grid's points to face in the initial frame. Using flexible models, the grid-tracking and deformed technique employed follows the grid in a series of video frames as the emotion changes continuously unless it reaches the frame with the strongest emotion. A unique multiple-class Support Vector Machine (SVM) framework of classifiers that operates to identify both the six fundamental facial gestures and an array of picked facial Action Units (FAUs) uses the geometrical movement of some chosen Candide nodes, which can be expressed as the variation among the node coordinates among the initial and highest emotion intensity frames. According to FAU recognition, the findings in the Cohn-Kanade database provide a recognition precision of 95.1% for FER and 99.7% for facial emotion identification utilizing the suggested multiclass SVMs.

Oh et al.[39] create a deep learning (DL) model for emotion classification using facial expressions that are resistant to distortion within the picture data. 53 participants were asked to express four distinct reactions for 15 seconds at a time; then, feelings were classified as positive or negative. After extracting facial emotions from the captured photographs and storing them as image data, noise was added in five stages to analyze the data set. Furthermore, when the noise was applied, its predictability was verified. This work verified that emotions might be identified from facial expressions using a DL model that is resistant to noise in a given data set.

Liu et al.[40] study the multimodal emotion integration process and companion robot empathic reactions. Natural HRI requires humanoid robots to be able to understand sympathetic facial reactions and respond accordingly. In order to create a multimodal emotion classification mechanism, this work integrates the GPT-3.5 system with a facial emotion-recognizing system. This work also tackles the problem of accurately replicating human facial expressions through the mechanical structure of a humanoid robot. The linked GPT-based multimodal emotion identification system employing text and images works better than existing approaches, according to studies conducted on benchmark databases, and has unique empathic reaction capabilities.

Kim et al.[41] suggest a new FER system design that depends on layered DL. In a system with hierarchy, the geometrical feature and the feature taken from the looks of the feature-based network are merged. The visual feature-based model uses the preprocessed picture to identify the holistic elements of the face, while the geometric feature-based network uses action unit landmarks to learn the position shift of a muscular system that moves primarily during facial emotions. With the help of this method, the extraction of variable facial characteristics is possible without the need for sequential information among the neutral and emotional photos.

Filippini et al.[42] suggest a highly efficient convolutional neural network (CNN)-based FER system that can be used to improve the understanding of the NAO robot of human facial reactions while providing it with the ability to determine the level of arousal of an interlocutor. In fact, the model evaluated in HRI was less successful in identifying neutral and furious emotions and 75% accurate in identifying shocked and afraid expressions. It was also 90% accurate in identifying happy and distressed facial expressions, respectively.

Martinez et al.[43] provide a facial emotion recognition technique that relies on the most recent ML techniques released for the state of the art and was built using the Robot Operating System (ROS), Noetic version. The system can identify emotions including rage, happiness, sadness, or shock. Using an inexpensive multipurpose board, the Raspberry Pi 4 Model B, comprehensive tests were carried out in a real-world setting to make these approaches more effective and to demonstrate that they could be implemented in real time using an inexpensive board. The whole FER system described in this paper can perhaps operate in real time with higher accuracy.

Alonso-Martín et al.[44] provide a multimodal approach for social robots to recognize users' emotions. The purpose of this framework is to facilitate HRI. It is a component of the Robotics Dialog System (RDS), which is the robot's whole communication system. The Gender and Emotion Facial Analysis (GEFA) technique was subsequently created to detect emotions in facial expressions. To establish the ultimate decision rule and evaluate the accuracy of every aspect, many tests were conducted with actual users. The results of the trials using this decision rule demonstrate a high accuracy rate in automatically identifying the emotions of the user, outperforming the results provided by the two sources of information (image and audio) separately.

Yan et al.[45] offer a multilayered CNN model utilizing a cross-attention mechanism (MCNN-CA) for precise emotion recognition. The proposed paradigm facilitates the automatic extraction of pertinent features from multimodal data and the integration of feature maps from various sources as modules for future emotion recognition. During the feature extraction phase, several CNNs were developed to extract essential information from multidimensional characteristics. The feature fusion module was employed to improve the intercorrelation among features via a channel-efficient attention method. This innovation demonstrates efficacy in integrating unique characteristics inside a singular mode and across other modes. The model was evaluated by EEG emotion recognition studies utilizing the SEED and SEED-IV datasets. Multimodal emotion studies, using EEG and textual data from the ZuCo dataset, assessed the efficacy of the proposed model. A comparative examination with contemporary studies demonstrates that the proposed model surpasses others in accuracy, F1, precision, and recall scores.

Aguang et al.[46] examine the application of deep learning for FER utilizing the Emognition dataset, which has ten target emotions: amusement, wonder, enthusiasm, liking, surprise, anger, contempt, fear, sorrow, and neutrality. A sequence of data preparation steps was executed to transform video data into photos and enhance the dataset. This paper presents CNN models developed via two methodologies: transfer learning (fine-tuning) utilizing previously trained models Inception-V3 and MobileNet-V2, and constructing models from the ground up employing the Taguchi method to identify an efficient combination of hyperparameter settings. The suggested model exhibited commendable performance across various testing procedures, achieving an accuracy of 96% and a mean F1-score of 0.95 on the simulated data.

The suggested CNN-based system presents a competitive and lightweight alternative, demonstrating robust performance on controlled datasets. Although leading models in the literature frequently utilize ensemble learning or transformer-based techniques to achieve accuracy exceeding 90% on datasets such as RAF-DB or AffectNet, these approaches incur heightened computational complexity and inference duration, rendering them less appropriate for real-time human–robot interaction contexts. Conversely, our model attains an accuracy of up to 95% on CK+ and has commendable performance (about 64%) on FER2013 without extensive preprocessing or ensemble methods. This equilibrium between precision and efficiency underscores the appropriateness of our methodology for implementation in real-time robots and assistive technologies. Moreover, in contrast to numerous opaque deep models, our research prioritizes interpretability by pinpointing the facial motions that significantly impact the CNN's decision-making, hence enhancing transparency in emotion inference. This study offers multiple significant contributions to the domains of HRI and computer vision-based FER:

- The study highlights the importance of facial expression detection in improving human-robot interactions, especially in areas such as surveillance, elderly support, and medical assistance. It examines how AI may enhance robotic systems' capacity to comprehend human emotional states via facial expressions.
- A digital image processing pipeline was developed and executed using computer vision and machine learning methodologies to identify human emotions based on facial expressions in photos and videos.
- The research effectively utilizes CNN architectures for the automatic extraction and classification of face emotions, illustrating the efficacy of deep learning in emotion identification tasks.
- A novel aspect of the study is the examination of which facial motions provide the greatest influence on CNN's emotion classification mechanism. This offers details on the interpretation and decision-making function of the overall system.
- The suggested model undergoes a thorough evaluation utilizing three prominent emotion datasets: FER2013, RAF-DB, and CK+. The model attains excellent accuracy on the CK+ data set (about 95%), indicating its trustworthiness in controlled settings, while exhibiting comparatively lower accuracy on FER2013 (around 64%), underscoring difficulties in real-world scenarios.
- The work enhances the awareness of dataset biases and the generalization ability of CNN-based FER frameworks by evaluating the proposed model's performance over datasets with differing complexity and noise levels.
- The results and the implementation strategy indicate practical relevance in real-time human-robot interaction systems, facilitating the development of more emotionally intelligent robots and assistive technologies.

Background information

FER may be applied in several domains, including interpersonal and linguistic communication, psychological clinics, psychology, neuroscience, pain assessment, detection of deception, intelligent environments, and multisensory HCI. As shown in Fig. 1, facial recognition can be performed using a variety of techniques, including feature-based techniques, neighboring graph techniques, expression-based techniques, and framework

approaches[47]. The form and texture of human faces are identical; therefore, a model acquired from an adequate number of faces could be used to recognize the other faces, which are not included in the training of the network process, with good generalization. A complex set of steps must be taken to develop the whole FER framework because the efficiency of the trained model is dependent on numerous factors, including the number of images in the training data set, the addition of data, the structure of CNN, hyperparameter adjustments, loss function, fine-tuning, transfer learning, and measurement metrics.

CNNs are an established class of deep learning algorithms utilized by computer vision, including automatic identification of emotions[48]. CNNs are made to autonomously recognize and identify features from pictures that might be used to predict the emotions of a person based on the picture[49,50]. CNNs might be trained using huge databases of pictures with matching emotional labeling in the framework of emotion identification to discover patterns and features linked to various emotions. Once trained, CNN might be utilized to predict new, unlabeled images by feeding them via the network and choosing the label that has the best chance of prediction. Next, a trained CNN might be used to predict the emotional impact of new images[51]. There are several benefits in using CNNs for emotion identification over more conventional techniques for image processing. For instance, CNNs might be trained to autonomously identify relevant characteristics from pictures instead of needing the tedious and mistake-susceptible process of human feature extraction. Furthermore, advances in DL have made it possible to handle different poses, brightness, and other environmental aspects that can alter the look of pictures, especially with the incorporation of approaches, including CNN and other supportive approaches[52].

CNNs are highly effective at solving complicated issues such as classification of images and analysis of medical images because they do away with the requirement to manually extract features. Initially, we observe a basic CNN template comprising many components that we may readily comprehend and associate with the suggested CNN model. A fundamental CNN has three sorts of layer: input, hidden, and output. After entering CNN along its input layers, the data pass through several hidden layers and eventually arrive at the output layer[53,54]. The output layer represents the network forecast. The output of the network is contrasted with the real labels to detect any error or loss. A foundational component of data processing is provided by the network's hidden layers. It is possible to break down each layer into its four sub-functions: convolutional layers, pooling layers, normalization, and activation. Multiple layers of neurons make up CNNs, and the most significant layers are the convolution layers and the pooling layers. Figure 3 shows the basic architecture of the CNN model and its implications for FER. The input image shows a happy emotion instance associated with a human emotional situation.

Convolutional layers

In a CNN, convolutional layers are primarily responsible for extracting features from the input images. Each convolutional layer contains a set of filters (or kernels) which are trainable parameters learned during the training process. Typically, the filters' size is less than the picture's real size. Every filter produces an activation map by convolving using the picture. A mathematical process known as convolution integrates two functions to explain their overlap. The filter is slid over the image's width and height during convolution, and at all geographical points, the product of dots among each filter component and the source data is computed. When two functions are combined, one is "slid" over the other, multiplied at all points of overlap, and the results are added to produce a novel function. Through this procedure, a novel function is created that shows the interactions between the two primary functions. Traditionally, convolution represents an integral, as shown in Eq. (1)[55], which could be used to indicate the degree to which one function, $p(t)$, overlaps another function, $q(t)$, when it is moved over the other.

$$(p * q)(t) \approx^{def} \int_{\infty}^{\infty} p(\tau)q(t-\tau)d\tau \tag{1}$$

During network training, the network learns these filter configurations. Usually, lower-level features are extracted from the information by the starting layers, and higher-level features are extracted by the deeper layers. Each

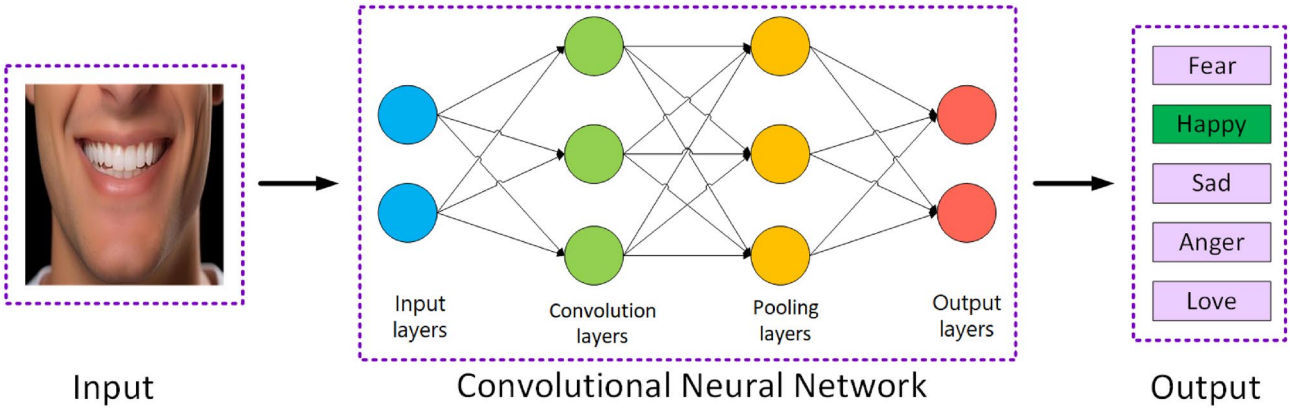

Fig. 3. Illustration of the basic architecture of the CNN-based FER technique.

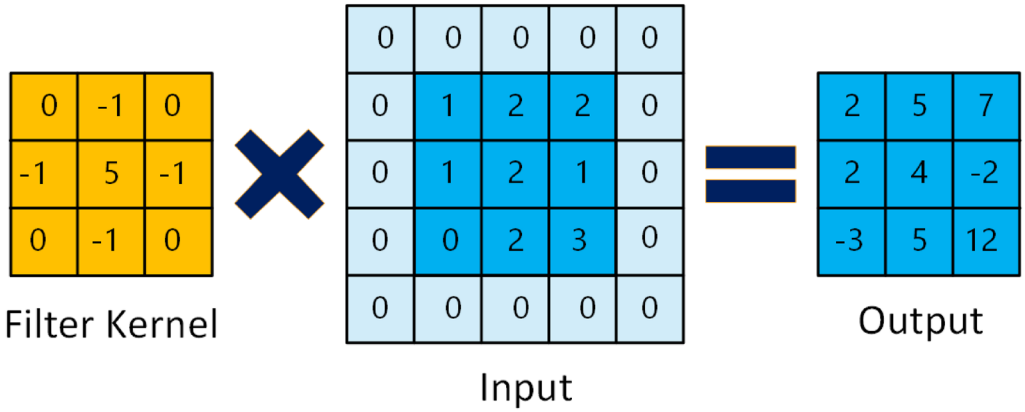

Fig. 4. Illustration of convolution operation using filter kernel.

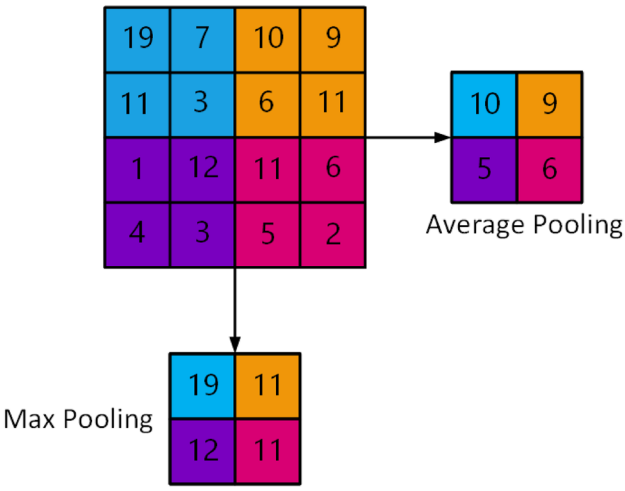

Fig. 5. Illustration of average pooling and max pooling calculation.

neuron in the map of features contains a receptive field that is coupled with its neighbors in the layer before it within a group of trainable weights that are sometimes thought of simply as a filter pool. The new feature map can be evaluated by convolving the input data with its learned weights, and then the convoluted results can be subsequently sent through a non-linear activation function[56]. Convolutional filtering might be implemented in image processing to perform techniques that include sharpening, blurring, and edge detection for any image. The right kernel, or convolution matrix, is chosen to perform this. Figure 4 illustrates the convolution operation with a 3×3 sharpening kernel filter.

Pooling layers

The pooling layer is a crucial element of CNNs. It is generally used in conjunction with the convolutional layer to decrease the spatial extent of feature maps and minimize both the quantity of data and the variable value. The pooling layer functions by dividing every feature map into predetermined areas and performing actions, including identifying the largest value (max pooling) or computing the mean value (average pooling) inside each area[57]. The two primary categories of pooling procedures are average pooling and max pooling. The max pooling is the largest score found in each zone, while the average pooling is the average score computed for each zone. Figure 5 shows the estimate of the average and max pooling. By exhibiting a certain level of translation invariance and aiding in the preservation of important properties, these processes strengthen the resistance of the network to even a small amount of changes in the input.

In CNNs, the pooling layer has two main functions. First, it shortens the learning process and minimizes computing complexity by decreasing the physical dimensions of the feature maps. Secondly, it improves the generality and interpretability of the framework by extracting important features and eliminating unnecessary information[58]. A stochastic method known as pooling enables the aggregation of the outputs at a designated location into a singular value. The pooling is an essential phase in convolutional methods to reduce the dimensionality of the extracted features. The pooling approach reduces the computational costs of the top layers

and provides supplementary spatially altered data by removing the imperfect interconnection present between the convolutional layers[59]. The average pooling and the max pooling can be determined using Eqs. (2) and (3)[60].

$$P_{avg} = \frac{1}{N}\sum_{j}^{N}(x_j) \tag{2}$$

$$P_{max}(X) = max_j(x_j) \tag{3}$$

Where a rectangular panel that has N permutations in a picture is represented by the vector x.

Fully-connected layers

A fully connected layer denotes a neural network configuration in which each neuron performs a linear transformation on the given input vector via a weight matrix. Consequently, all potential layer-to-layer linkages exist, indicating that each element on the input vector affects each component of the output vector. In classification tasks, fully connected layers convert high-level information into scores, which are usually processed using a softmax function to obtain probabilistic class forecasts.

ReLU activation function

A well-liked activation function that is commonly utilized in deep learning models is the Rectified Linear Unit (ReLU) activation function. The ReLU activation function can be used to give the framework as a whole non-linearity without affecting the reception fields of the convolutional layer. The function is a simple but powerful tool for modeling neural networks since it yields a result that is exactly proportionate to the input. Whenever the input is positive, the ReLU function provides the value directly; if not, the result is zero. Because it solves the problem of disappearing gradients and enables models to train more quickly and efficiently, it has been the standard activation function in a variety of neural networks[61]. The ReLU function can be represented as Eq. (4)[62]. In addition, the ReLU function can be graphically represented as in Fig. 6.

$$R_p = max(0, p) \tag{4}$$

where p represents the specified pixel value and $R(p)$ represents the ReLU activation function.

Simulation and experimental analysis

It has demonstrated significant efficacy in determining and obtaining patterns from visual input, which makes it particularly suitable for applications, including recognizing objects and classifying images. CNNs differ from multilayer perceptrons (MLPs) in that they possess hidden layers known as convolutional layers. The proposed approach relies on a multilevel CNN structure. The proposed study uses a CNN for FER tasks, leveraging 48x48 grayscale images as input for both the FER2013 and CK+ datasets. Additionally, the RAF-DB dataset, which measures 100 × 100, incorporates colored graphics. The model includes Conv2D layers that work quickly and reduce the number of variables while still effectively extracting important features. The design has four groups of convolutional layers, with each group having two convolution layers, followed by batch normalization, ReLU activation, MaxPooling, and dropout to reduce overfitting. The network incrementally improves the filter count from 64 to 512, allowing it to distinguish complex face behaviors. After the convolutional layers, a Flatten layer is added, then a fully connected dense layer with 512 neurons, and finally a softmax layer that has 7 outputs for the 7 basic emotions. We train the model using categorical cross-entropy loss and an optimizer, such as Adam, which enhances its performance on the FER2013, RAF-DB, and CK+ datasets. The initial suggested level provides background removal, utilized to extract emotions using a picture[63]. This work used FER2013, RAF-DB, and CK+ datasets to train the CNN algorithm suggested for the recognition of human facial expressions. The proposed model has a sequential structure. A substantial data set of annotated facial expressions is necessary

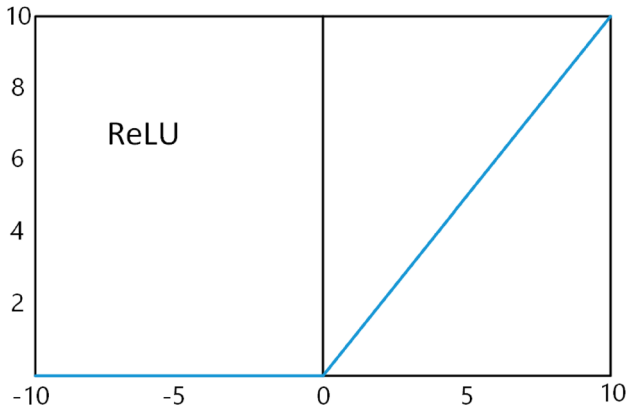

Fig. 6. Illustration of ReLU activation function.

Dataset	Number of samples
FER2013	32,298
RAF-D	15,339
CK+	981

Table 1. Illustration of the number of samples of data in different datasets.

Type of dataset	FER2013	RAF-DB	CK+
Training set	28,709	12,271	783
Test set	3589	3068	198

Table 2. Illustration of training and test datasets.

System configuration	Descriptions
Python version	3.12.7
TensorFlow version	2.18.0
Numpy version	1.26.4
Learning rate	0.001
Optimizer	Adam
RAM	32GB
Processor	11th Gen Intel(R) Core(TM) i7-11800H @ 2.30GHz

Table 3. List of hyperparameters.

for the training of the proposed CNN model for FER tasks. We analyze the FER2013[64], RAF-DB[65], and CK+[66] data sets during training and validation of the proposed model, which are publicly available. FER2013 comprises more than 30,000 RGB facial pictures exhibiting various emotions, each constrained to a dimension of 48×48 pixels, with the primary labels classified into seven distinct types: 0 = angry, 1 = disgust, 2 = fear, 3 = happy, 4 = sad, 5 = surprise, and 6 = neutral. The faces are automatically aligned to ensure that they are about centered and occupy the same amount of space across every image. RAF-DB is an extensive facial emotion dataset of over 30,000 varied facial photos sourced from the Internet, each constrained to a dimension of 100 × 100 pixels, with the primary labels classified into seven distinct types. Each picture has been individually annotated by around 40 contributors, according to the crowd-sourcing methodology. The images in this database exhibit significant variations in participants' age, gender, ethnicity, head postures, lighting circumstances, occlusions (e.g., spectacles, facial hair, or self-occlusion), and post-processing techniques (e.g., multiple filters and specialized effects). RAF-DB has significant diversity, substantial volumes, and comprehensive annotations. CK+ is a small data set that contains 981 total images of various types of emotional image, each constrained to a dimension of 48×48 pixels, with the primary labels classified into eight distinct types. Tables 1 and 2 show the complete description of the FER2013, RAF-DB, and CK+ datasets, where Table 1 illustrates the amount of sample data for each emotion, and Table 2 illustrates the amount of sample data for training and test datasets.

The typical CNN network unit is used to retrieve the main expression vector. There are three steps in the FER method. Firstly, the dataset will pass through preprocessing to remove unwanted information and make the dataset format according to the required shape. The updated format will operate and yield effective results based on a broader technique. In the second step, feature extraction uses real-time image collection to identify the face in the shape of a facial frame. Lastly, the emotion detection step involves using the CNN algorithm to classify the image within one of seven categories. Furthermore, a schematic diagram is used to illustrate these key stages, as shown in Fig. 2. The FER test is performed in situations where the face is inclined to the right or left, looking upward or below, and spinning left or right. The objective of the test is to focus on the results of the image acquired by the camera. The technology can distinguish facial expressions depending on the camera's observation angle. Table 3 contains the hyperparameters and system configuration information that are used first for training and then for validating the proposed FER framework.

The public data sets of FER2013, RAF-DB, and CK+ are used to assess the generalizability and reliability of the presented FER model. This work provides excellent information on the use of CNN for the emotion classification process using different FER datasets. We applied a native Python script to access the open data sets FER2013, RAF-DB, and CK+, and, additionally, the data were preprocessed to eliminate unnecessary data without missing important details. CNN sequence modeling is acknowledged for its capacity to model, evaluate, forecast, or generate sequential data. Regular recording of the input information takes place, and a common method for segmenting the resultant information from the sensors remains to use an adjustable window with a predetermined size. The convolutional layer has weights that require training, whereas the pooling layers employ a predetermined function to transform activation. Despite contemporary advances in image resolution,

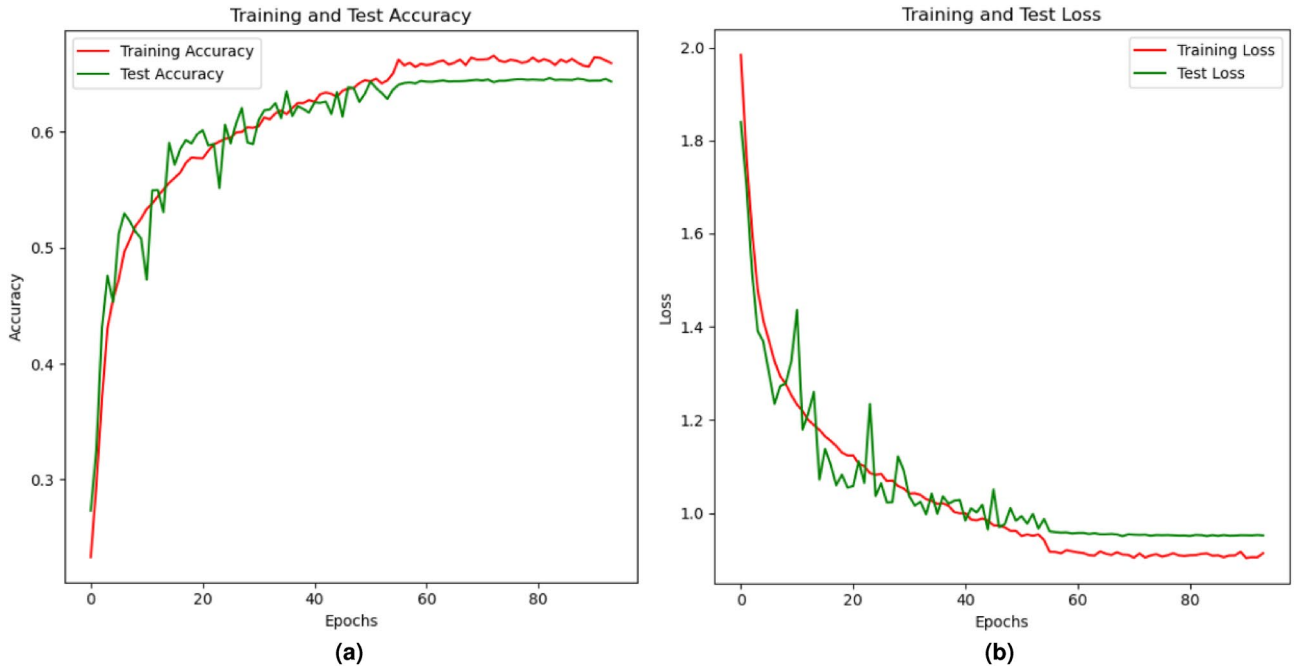

Fig. 7. Illustration of training and test estimation for the FER2013 dataset, where (**a**) training and test accuracy and (**b**) training and test losses.

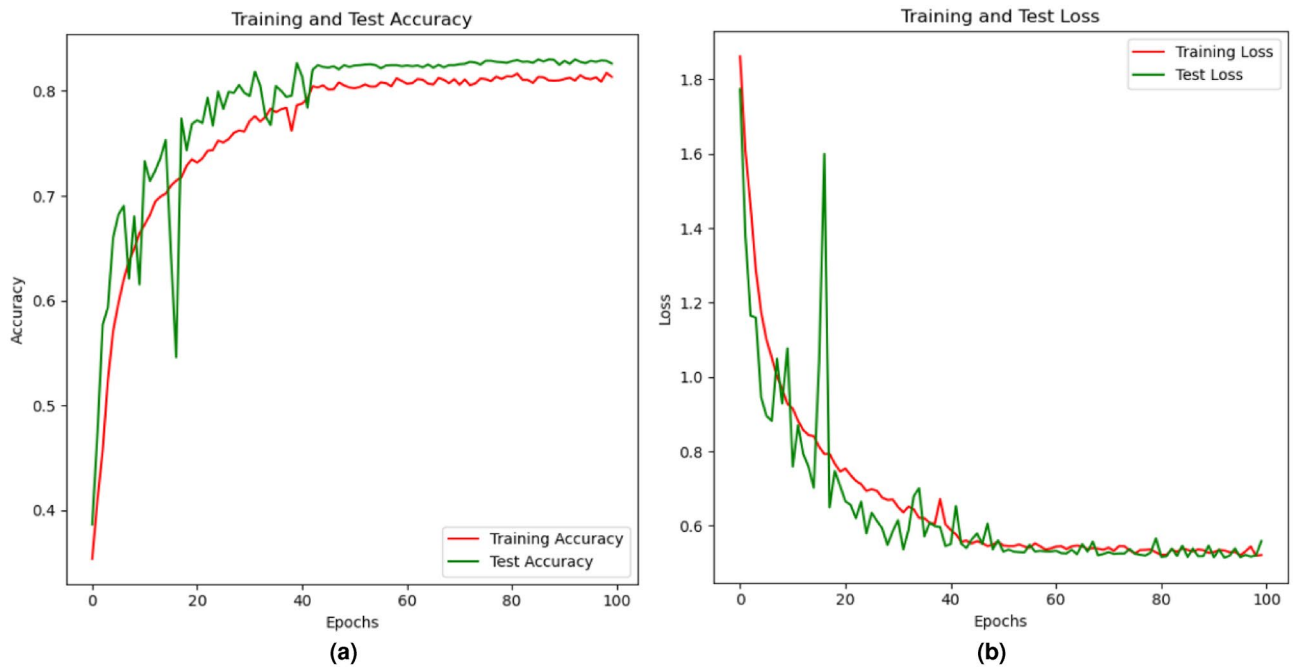

Fig. 8. Illustration of training and test estimation for the RAF-DB dataset, where (**a**) training and test accuracy and (**b**) training and test losses.

reducing the size of images renders them more challenging to identify. However, poor resolution significantly accelerates the training cycle with minimal effect on accuracy. Only zero-mean normalization is required for data preprocessing in this context. Figure 7 shows the accuracy of the training and test and the loss of training and test during the simulation for the FER2013 dataset. Figure 8 shows the training and test estimation for the RAF-DB dataset during the simulation. Figure 9 shows the training and test estimation for the CK+ dataset during simulation.

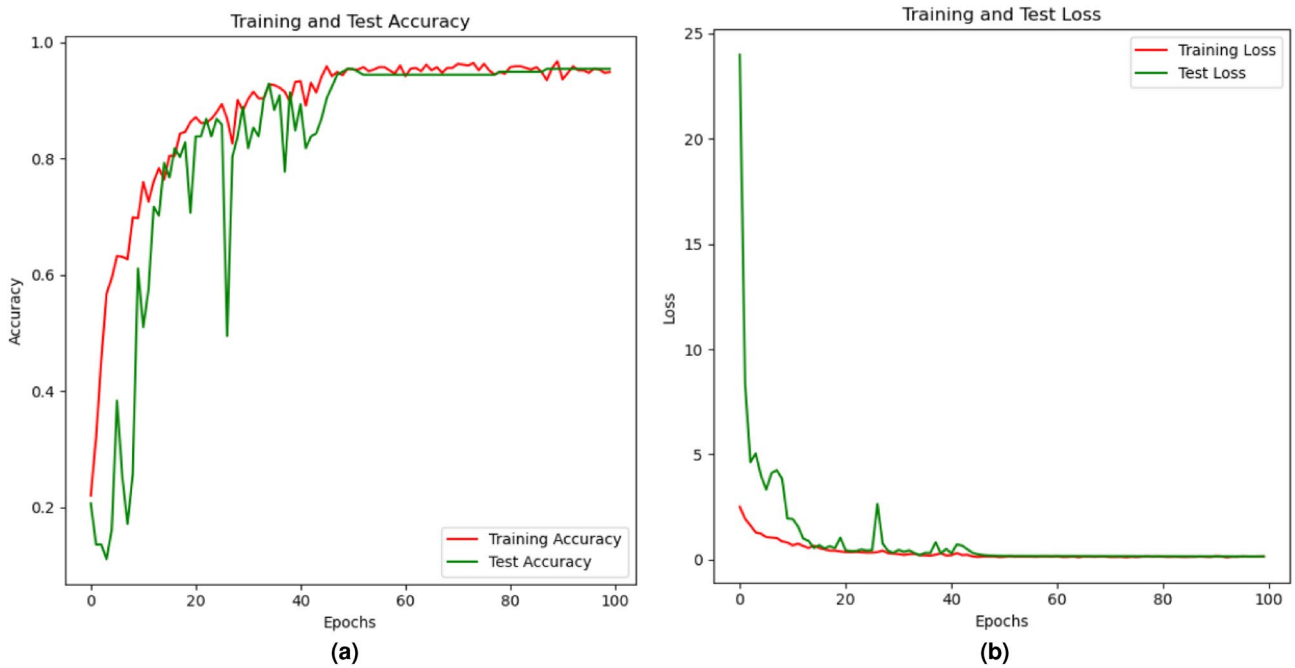

Fig. 9. Illustration of training and test estimation for the CK+ dataset, where (**a**) training and test accuracy and (**b**) training and test losses.

This code presents an algorithm based on deep learning designed to recognize emotions on faces accurately in images. The foundation of this facial emotion recognition system consists of a CNN architecture developed using the TensorFlow and Keras packages. We aimed to assess the model's ability to recognize diverse facial gestures by training it on a comprehensive collection of images and subsequently evaluating its performance with that same dataset. CNNs are distinguished by their enhanced depth, attained by the sequential stacking of many convolutional layers. The suggested CNN approach for FER employs many deep learning techniques, such as data augmentation and learning rate reduction. These techniques augment the model's capacity for generalization to novel data and mitigate overfitting. The proposed work has the potential to achieve optimal results using the FER2013, RAF-DB, and CK+ datasets. The values of each pixel in the picture are normalized to a range between 0 and 1, a common pre-processing procedure for the image data. The process facilitates a reduction in the complexity of the model and accelerates training. Next, we apply data augmentation techniques to the training data, thus enhancing its diversity and adaptability to various types of images. We particularly rotate, modify the dimension and size, shear, enlarge, and horizontally flip the photographs. Throughout training, various augmentations are arbitrarily implemented on each picture gathering, producing a novel array of images dynamically for each epoch.

The system will comprise four primary components: data collection, data pre-processing, deep learning models, and facial emotion recognition (FER) results. The data-gathering component will collect a dataset of facial images from live webcams. This data set will train deep learning models for FER and restore images with the FER 2013 dataset. CNNs will be used to construct deep learning frameworks for FER. We suggest that by using the complete datasets FER2013, RAF-DB, and CK+, improving data quality through cleaning techniques, and carefully addressing class imbalance, we can develop a new deep CNN model that performs better than existing ones. CNNs require adequate training data to ensure adaptability for a specific recognition task. However, most publicly available data sets for facial emotion recognition lack a sufficient number of photos for training purposes. Consequently, data augmentation is an essential process for deep FER. Data augmentation methods may be categorized into two types: spontaneous data augmentation and offline data augmentation. The preliminary stage entails the purification of the dataset and the rectification of labels for specific occurrences to produce a revised version of the FER2013, RAF-DB, and CK+ datasets. Subsequently, we focused on examining the impact of various balancing procedures on this dataset. Moreover, FER 2013 constitutes a substantial dataset; training and testing procedures were executed using the datasets it offers. To mitigate overfitting, we meticulously selected the optimal weight according to the system's learning trajectory to get sufficient performance. In addition, intensity normalization was employed to increase the contrast in face photographs. A data augmentation method was used to address the limited quantity of training photos in the FER challenge.

Fluctuations in lighting and head orientations can significantly alter pictures, compromising the ability to recognize facial emotions. In contrast to traditional approaches, which treat feature extraction and classification as separate processes, deep networks might perform facial emotion recognition in an end-to-end manner, effectively addressing the challenge of lighting variations. A loss layer is incorporated at the end of the proposed network to manage the backpropagation error, allowing the network to quickly output the prediction rate for

all samples. In CNNs, the softmax function is predominantly utilized to reduce the cross-entropy between the predicted class probability and the actual distribution. The ROC curve assesses the efficiency of a recognition method. The ROC curve displays the true positive rate (TPR) and false positive rate (FPR) at various threshold values. The TPR denotes the ratio of accurately recreated faces in face restoration, whereas the FPR denotes the ratio of inaccurately recreated faces. The AUC can be presented to illustrate the overall accuracy of the model. The ROC curve can be useful for evaluating the efficiency of the models and comparing them with alternative models. The spectrum for all emotion classifications is notably significant. ROC curves and the associated area under the curve (AUC) for each individual facial emotion recognition outcome illustrate recognition efficiency. Figure 10 illustrates the receiver operating characteristic (ROC) curve for all expression recognition results for the FER2013, RAF-DB, and CK+ datasets. Where, for the FER2013 dataset, an increase in ROC indicates an improved ability of the framework to distinguish between distinct classes. The values of the disgust, happiness, and neutral labels are 95%, 97%, and 97%, respectively. The others achieved commendable percentages of 89%, 83%, 90%, and 86% for the respective emotional categories of anger, fear, sadness, and surprise. For the RAF-DB dataset, an increase in ROC indicates an improved ability of the framework to distinguish between distinct classes. The values of the angry, sad, happy, and surprise labels are 98%, 98%, 98%, and 98%, respectively. The others achieved commendable percentages of 95%, 91%, 96% for the respective emotional categories of disgust, fear, and neutral. For the CK+ data set, an increase in ROC indicates an improved ability of the framework to distinguish between distinct classes. The values of the labels fear, happy, anger, sadness, and surprise are 100% for all, respectively. In addition, the contempt label is 99%, so the proposed technique has the highest performance for the CK+ dataset.

The findings demonstrate that the framework adequately anticipated the emotions of several elements observable in the live video feed. The constructed model was capable of identifying emotions such as happiness, sadness, fear, anger, disgust, surprise, and neutrality for the FER2013, RAF-DB, and CK+ datasets. The created model has potential uses in HRI, marketing studies, and psychological diagnostics. The method we used has been trained for 100 epochs, achieving the test accuracy of 64% for the FER2013 dataset, the test accuracy of 83% for the RAF-DB dataset, and the test accuracy of 95% for the CK+ dataset. Figure 11 illustrates the confusion matrix for the CNN model. This confusion matrix illustrates the effectiveness of a multi-class FER technique. Each row denotes the actual emotion label, whereas each column signifies the anticipated label; the intensity of the cell's color indicates the number of instances it encompasses. In the FER2013 dataset, the confusion matrix indicates that the approach performs well in identifying happy, sad, and neutral emotions, as demonstrated by the large number of right forecasts across the diagonal. However, it experiences difficulties with fear, surprise, and disgust, frequently confusing them with other analogous emotions such as anger or sadness. The model has greater accuracy with clear emotions but struggles to differentiate between complex or overlapped visuals. The confusion matrix indicates enhanced classification performance, particularly for Happy, Sad, and Neutral, with substantial right forecasts (for instance, 1068 for Happy, 408 for Sadness, and 556 for Neutral). All emotions show a marked reduction in misclassifications compared to the FER2013 confusion matrix, indicating an overall improvement in model accuracy. However, slight ambiguity persists in some situations, especially with neutral being incorrectly classified as sadness or fear, and fear being incorrectly classified as neutral or surprise. This confusion matrix shows great classification performance, with nearly all predictions arranged on the diagonal, which means that most emotions are correctly identified. Every emotion category (such as happiness, surprise, fear) exhibits minimal misclassifications, with only slight misunderstandings, such as three 'disgust' instances misclassified as 'anger' and one 'surprise' instance as 'contempt.' This evidence indicates that the model is very precise and is properly tuned for the FER task. Table 4 provides the classification report of the proposed model for different facial emotions in the FER2013, RAF-DB, and CK+ datasets using precision, recall, and F1 scores. The classification report illustrates the performance of an emotion recognition model among three datasets: FER2013, RAF-DB, and CK+. The CK+ data set generates higher scores for all emotions, indicating that the model exhibits optimal performance in this excellent controlled data set. The RAF-DB data set has acceptable results in general, particularly for emotions such as happiness, anger, and sadness; however, it encounters greater difficulties with fear and contempt. However, FER2013 exhibits the poorest performance, especially for fear and disgust, probably because of its inclusion of more complex real-world imagery. This comparison underscores the significant impact of the quality and conditions of the data set on the reliability of emotion recognition.

The proposed approach is compared with state-of-the-art techniques and shows that the proposed approach has a better accuracy rate. The detailed comparative analysis is provided in Table 5. For the FER2013 dataset, the suggested technique achieves 64.00%, narrowly surpassing previous models, indicating a minor enhancement in a demanding dataset. The method suggested on the RAF-DB dataset reaches 83.00%, which is much better than other methods, showing that it works well with real images. The proposed framework achieves 95.00% on CK+, surpassing present techniques and indicating outstanding performance on controlled datasets. The suggested technique shows steady improvements on all datasets, especially RAF-DB and CK+, highlighting its effectiveness and superiority over state-of-the-art CNN-based systems.

Challenges and future research trends

The most widely used method for image analysis is CNN. Future endeavors should concentrate on achieving enhanced precision in the classification of a user's facial expressions. Providing information on observed facial expressions is essential for an accurate categorization of emotions. Research and design in HRI require much increased involvement from the human factors community compared to previous instances, with the exception of certain domains like commercial aviation and defense systems, where human factors specialists have historically participated. Modern advances in autonomous vehicles and drones have substantial obstacles in terms of safety and social acceptance[76]. All of the modalities analyzed offer potential sources of knowledge for future research and development. For example, the quick development of emotionally aware robots is expected

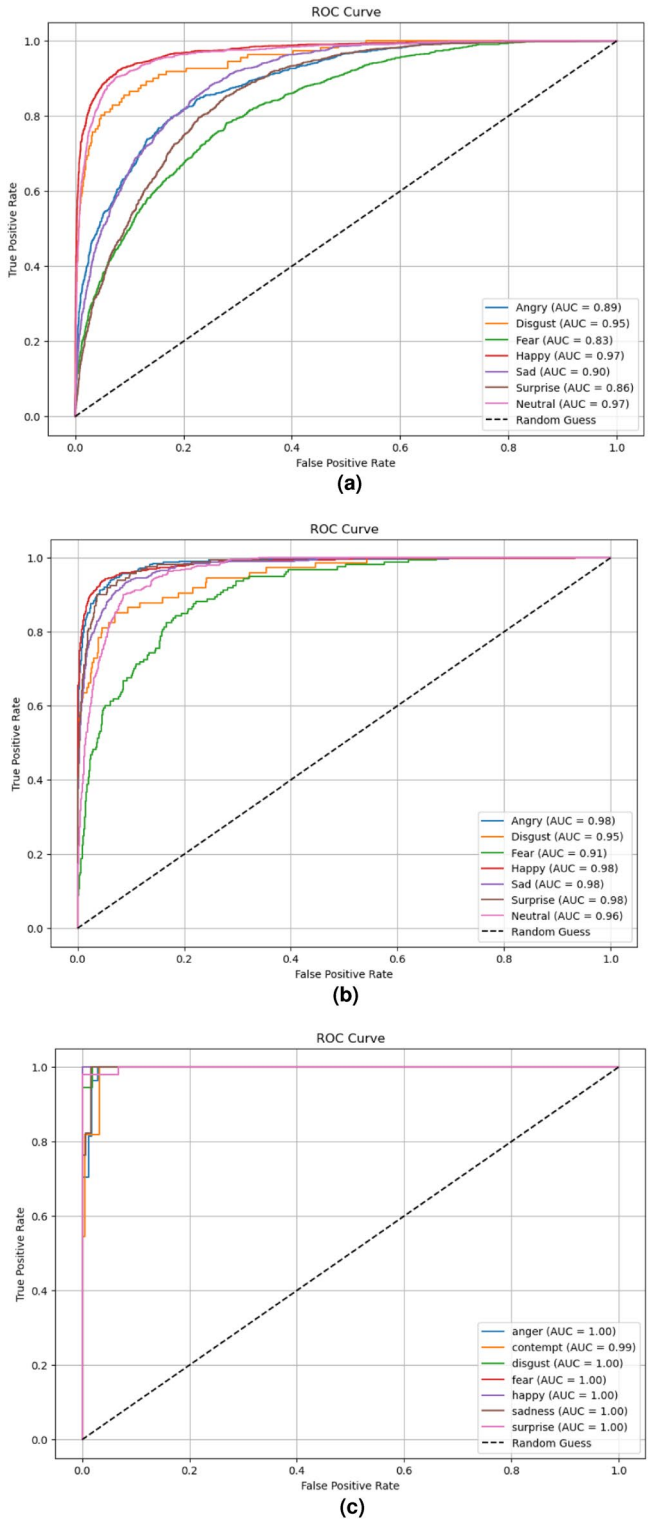

Fig. 10. An illustration of the ROC curves for different types of FER datasets.

to result from the combination of ML advancements with accessible and inventive technology, including depth sensors and smart gadgets. However, the demand for reliable findings, a trustworthy relationship, and the time limitations necessary to incorporate the observed emotion within the modification of robotic behavior makes the identification of emotions with robots a challenge today. The goal of future research might be to categorize facial reactions of users with greater accuracy. True emotion classification requires an understanding of the recognized facial gestures; it is suggested that larger, higher-resolution, higher-dimensional images be used and that more users participate in the video evaluation to see whether improved outcomes in FER might be

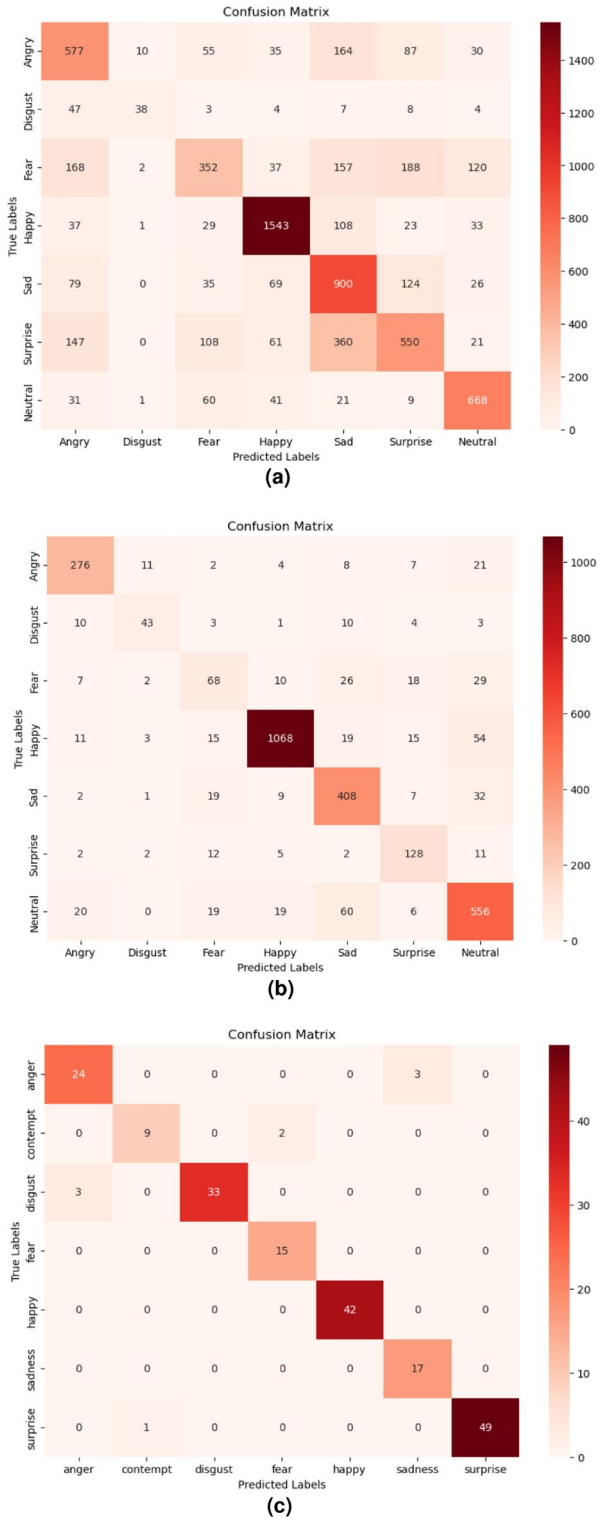

Fig. 11. An illustration of the confusion matrix for different FER datasets during simulation.

achieved. A temporal series of facial expressions might be used to train a new emotion classifier that uses long-short-term memory (LSTM) networks[77]. A time series can deliver the categorization with some context, since it can be determined by an array of images that define a gesture of an emotion rather than just one specific facial expression. Future advancements will use many inputs, including infrared photos, audio models, and physiological information and depth information collected from 3D facial models, to increase the accuracy of methodologies[78]. The HRI model created in this study will be improved for use in medicine, primarily to help elderly patients suffering from dementia and to improve the interpersonal skills of kids diagnosed with autism

Emotion	FER2013			RAF-DB			CK+		
	Precision	Recall	F1	Precision	Recall	F1	Precision	Recall	F1
Anger	0.53	0.60	0.56	0.84	0.84	0.84	0.89	0.89	0.89
Disgust	0.73	0.34	0.47	0.69	0.58	0.63	1.00	0.92	0.96
Fear	0.55	0.34	0.42	0.49	0.42	0.46	0.88	1.00	0.94
Happy	0.86	0.87	0.87	0.96	0.90	0.93	1.00	1.00	1.00
Sadness	0.52	0.73	0.61	0.77	0.85	0.81	0.85	1.00	0.92
Surprise	0.56	0.44	0.49	0.69	0.79	0.74	1.00	0.98	0.99
Neutral/Contempt	0.74	0.80	0.79	0.65	0.82	0.80	0.90	0.82	0.86

Table 4. Classification report for different facial emotions on FER2013, RAF-DB, and CK+ Datasets.

Reference	Type of dataset	Type of model	Accuracy rate
Diaz et al.[67]	FER2013	CNN	63.00%
Liu et al.[68]	FER2013	CNN	61.74%
Gursesli et al.[69]	FER2013	CNN	63.00%
Proposed technique	FER2013	CNN	64.00%
Ullah et al.[70]	RAF-DB	CNN	65.4%
Lin et al.[71]	RAF-DB	CNN	75.73%
Li et al.[72]	RAF-DB	CNN	74.00%
Proposed technique	RAF-DB	CNN	83.00%
Zhang et al.[73]	CK+	CNN	92.35%
Jain et al.[74]	CK+	CNN	92.71%
Lopes et al.[75]	CK+	CNN	92.73%
Proposed technique	CK+	CNN	95.00%

Table 5. Comparative estimation of the proposed technique with state-of-the-art techniques.

spectrum disorder. Through interaction with a humanoid robot, medical input from psychologists, doctors, instructors, and therapists might be used in the research to alleviate anxiety and dissatisfaction.

Conclusions

This paper presents an image-driven computer vision methodology for advancing CNN techniques for performing FER using different datasets. The proposed framework is suitable for real-time recognition of emotions in many applications, including augmented reality, HRI, and robots. Advanced approaches, including data augmentation, transfer learning, and optimizing hyperparameters, can help improve the FER framework even more. This work proposes an implementation of a framework capable of predicting and identifying facial emotion via the use of the CNN technique and methods to extract features. The proposed approach has great potential for FER and can make major contributions to AI and computer vision. A significant problem in previous studies was their reliance on static pictures that do not evolve over time, rendering them time-consuming and unsuitable for real-time applications. In contrast, our method can immediately forecast people's emotions using different FER datasets, including FER2013, RAF-DB, and CK+. These findings have substantial implications for FER since they illustrate the ability of ML algorithms to analyze facial expressions with precision and efficiency in real time. Employing advanced methodologies, including transfer learning, data augmentation, and hyperparameter tuning, can further enhance the model. The suggested model exhibits substantial potential for emotion recognition and could significantly advance the fields of computer vision and machine learning.

Data availability

Correspondence and materials requests should be addressed to the corresponding author on reasonable request. The data sets analyzed during the current study are publicly available in the Kaggle repository. Links to data sets analyzed during this study: RAF-DB dataset (https://www.kaggle.com/datasets/shuvoalok/raf-db-dataset), FER2013 dataset (https://www.kaggle.com/datasets/msambare/fer2013/fer2013), and CK+ dataset (https://www.kaggle.com/datasets/shuvoalok/ck-dataset/ck-dataset).

Code availability

The computer code used to support the findings of this study are made publicly available at: https://github.com/raj91ravi/Facial-Emotion-Recognition.

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Declarations

Competing interests

The authors declare no competing interests.

Additional information

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